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Subject : System design Assignment

Title: Food Delivery System Design (End-to-End with Queue & Pub/Sub)

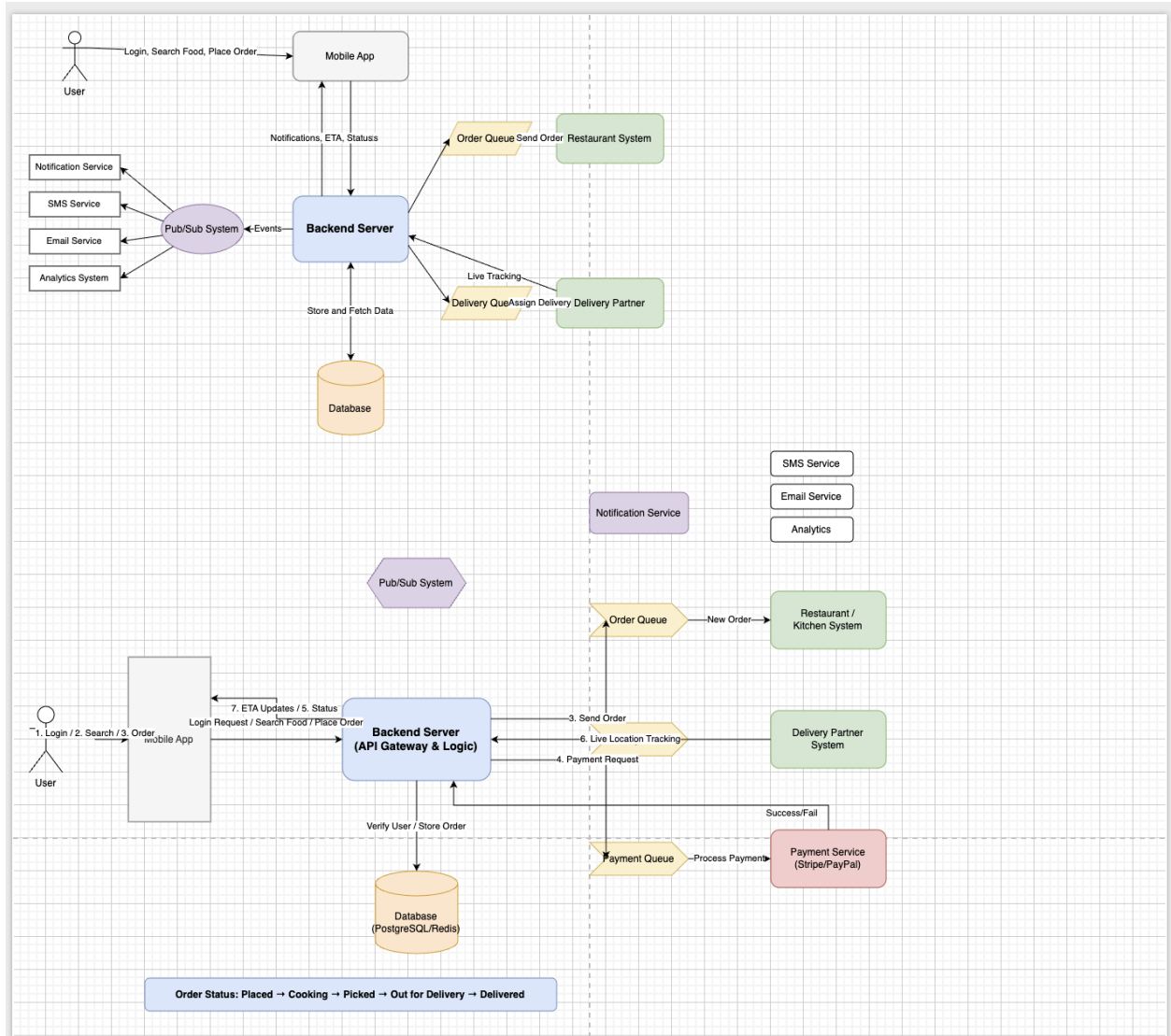
Task 1: End-to-End System Design (Food Delivery App)

Description :

This diagram shows how a food delivery app works. The user logs in, searches for food, and places an order through the mobile app.

The backend server processes the request, stores data in the database, and sends the order to the restaurant. A delivery partner is assigned to deliver the order.

The user receives updates like order status, live tracking, and ETA.



Task 2: Advanced System Design with Queue and Pub/Sub

This diagram shows the advanced design of a food delivery system using queue and pub/sub architecture.

Queues are used to manage multiple requests such as order processing, kitchen operations, payment handling, and delivery assignment. This helps the system handle high traffic efficiently.

The pub/sub system is used to trigger multiple services at the same time, including notifications, SMS, email, and analytics.

The diagram also shows real-world scenarios like high traffic handling, restaurant overload, and payment failure, making the system scalable and reliable.

